

WORK EXPERIENCE

Boston University | Boston, MA
Graduate Research Assistant

October 2024 - Present

- Implemented neural networks on 10 ETF datasets in Python at Shared Source Computing, achieving a 15% accuracy improvement over traditional models.
- Evaluated 3 distinct price intervals (close-to-close, open-to-close, close-to-open), identifying LSTM as the best-performing model with 8% higher accuracy compared to CNN and RNN.
- Used various data visualization techniques to deliver actionable insights on investment timing strategies, contributing to a 10% improvement in prediction accuracy for stock price trends.

TaskBench | Dubai, United Arab Emirates
Business Analyst Intern

June 2024 - August 2024

- Created detailed reports and business proposals for companies like UBM, Ahalia Hospital, Pharma Sol, and EAIC, leading to business expansion, successful collaborations, and securing funding worth \$400,000.
- Built a website for a company, enhancing their online presence and customer engagement by 32%.
- Managed 10+ B2B interactions, developing business model canvases for various projects that improved operational efficiency by 20%.
- Utilized Generative AI for document creation and analysis, and conducted in-depth audit reviews using Machine Learning models for financial companies.

Indian Institute of Management | Kerala, India
Research Intern

May 2022 - July 2022

- Collected and organized qualitative data from 50+ interviews and surveys on women from different societal segments.
- Utilized NVivo for systematic coding and thematic analysis of the qualitative data, ensuring accurate data categorization.
- Visualized data using Tableau, presenting insights that led to a 17% improvement in the research project's data analysis efficiency.

EDUCATION

Boston University | Boston, MA

September 2024 - December 2025

Master of Science in Applied Data Analytics **CGPA: 4.0/4.0**

Coursework: Data Science with python, Machine Learning, Data Analytics and Visualization

Vellore Institute of Technology | Chennai, India

July 2019 - August 2024

Integrated MTech in Software Engineering **CGPA: 9.09/10**

Coursework: Data Mining, Database Management, Big Data Analytics, Statistics

PROJECTS

Real-Time Market Data Aggregation and Analysis | Course

January 2025 - Present

- Designed and implemented real-time data dashboards using company API and tools like Tableau and Power BI, reducing market trend analysis time by 45%, enabling faster business decision-making.
- Optimized SQL queries and indexing strategies to improve database response times by 50%, supporting the aggregation and analysis of market data streams.
- Developed ETL pipelines that transformed and standardized 1TB+ of raw market data daily, achieving a data accuracy improvement of 30% through rigorous cleaning and validation processes.

Data-Driven Journalism: Extracting Insights from News Headlines | Course

September 2024 - Present

- Conducted sentiment analysis, topic modeling, and clustering on news headlines from major outlets using web scraping in Python and R.
- Performed statistical tests and comparative analysis to identify significant differences in sentiment, and media bias across sources.
- Developed visualizations to effectively communicate findings on sentiment distribution and editorial patterns.

Analysis of Crimes Against Children in India using Machine Learning Models | Course

July 2021 - December 2023

- Applied machine learning techniques (SVM, Random Forest, K-NN) on a dataset of 300,000+ crime records to classify and predict crime intensities, focusing on crimes against children in various regions of India. Utilized Hadoop and Hive to optimize data management, reducing retrieval time by 25%.
- Leveraged Tableau for visualizations (e.g., heatmaps, time series) to reveal crime trends and implemented the Apriori algorithm, uncovering 100+ associations and 5 key crime correlations across 30+ regions for deeper insights into crime patterns.

SKILLS

Programming Languages : R, Python, MATLAB, SQL, C

Tools/Software: Microsoft Office Suite, Tableau, Nvivo, Power BI, Alteryx, Jupyter Notebook, Visual Studio Code, Anaconda

Technologies/Frameworks: Keras, TensorFlow, PyTorch, Scikit-learn, Matplotlib, NumPy, Pandas, OpenCV, Hive, Hadoop

Soft Skills: Leadership, Verbal Communication Skills, Detail-Oriented, Problem-Solving, Critical Thinking, Business Accumen

PUBLICATIONS

- T. Kundu** and A. S., "Enhancing Deep Neural Network Architecture in Spatio- Temporal Forecasting Through Neural Architecture Search," 2024 International Conference on Recent Advances in Electrical, Electronics, Ubiquitous Communication, and Computational Intelligence (RAEEUCCI), Chennai, India, 2024, pp. 1-5, doi: 10.1109/RAEEUCCI61380.2024.10547787.
- T. Kundu**, J. S. V., S. B and P. B., "Analysis of Crimes Against Children in India Using Machine Learning Models," 2023 6th International Conference on Recent Trends in Advance Computing (ICRTAC), Chennai, India, 2023, pp. 78-85, doi:10.1109/ICRTAC59277.2023.10480723